

llmXive Automated Review of MCompassRAG: Topic Metadata as a Semantic Compass for Paragraph-Level Retrieval

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer’s exact instructions are public in the [llmXive agent-prompt library](#); each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer’s section also names the underlying model that produced it.

This paper’s panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_claim_accuracy.md

The paper makes several strong quantitative claims regarding Information Efficiency (IE) improvements and latency reductions that require precise alignment with the provided data tables.

First, the Abstract and Introduction claim an average IE improvement of **8.24%** over the “strongest non-LLM baseline.” In Table 1 (Main Results), the “LLM + 10 Topics” row (IE 40.83 on Dragonball) outperforms the proposed “MCompassRAG + 10 Topics” (IE 38.97). The text implies MCompassRAG is the best non-LLM method, but the “LLM + 10 Topics” row appears to be an oracle or a specific configuration of an LLM baseline. The claim of “8.24% improvement” must be explicitly tied to a specific baseline (e.g., SAKI-RAG with IE 32.90) to be accurate. If the comparison is against SAKI-RAG, the text should state “over SAKI-RAG” rather than the ambiguous “strongest non-LLM baseline,” especially since the “LLM + 10 Topics” row exists in the same table and outperforms the proposed method.

Second, the methodology section (Section 3.3) and Experimental Setup cite **Qwen3-Embedding-4B** and **Qwen3-32B**. As of the current public knowledge cutoff, the Qwen3 series has not been released; the latest public versions are Qwen2.5. Citing unreleased models as the primary student/teacher encoders without a clear note that these are internal or future models undermines the reproducibility of the results. If these are typos for Qwen2.5, the text must be corrected immediately. If they are internal models, the paper must clarify their availability status.

Third, the claim that MCompassRAG “closely approaches the LLM + 10 Topics oracle” is supported by the data (e.g., 94.13 vs 94.67 on SCI-DOCS in the text, though Table 1 shows 94.10 vs 95.83). However, the table labels the “LLM + 10 Topics” row as a baseline, not an oracle. The distinction between a standard LLM baseline and this “oracle” configuration is blurred. The authors should clarify if “LLM + 10 Topics” is a standard baseline they ran or a theoretical upper bound. If it is a standard baseline, the claim that MCompassRAG is “non-LLM” while being compared to an “LLM + 10 Topics” baseline needs careful phrasing to avoid confusion about whether the topic metadata is part of the LLM baseline or the proposed method.

Finally, the latency claim of “over 5x lower latency than LLM-based baselines” is supported by Table 2 (174ms vs 925ms for SAKI-RAG, though SAKI is not strictly LLM-based in the same way as PageIndex). The comparison to PageIndex (4408ms) is valid, but the “5x” claim should be explicitly linked to the specific LLM-based baselines (e.g., PageIndex, REFRAG) to ensure the comparison is fair and accurate. The current phrasing “strongest efficient RAG baselines” is slightly ambiguous given the mix of LLM and non-LLM methods in the tables.

Action items.

- **[writing]** The abstract and introduction claim an average Information Efficiency (IE)

improvement of 8.24% over the ‘strongest non-LLM baseline’. However, Table 1 shows MCompassRAG (IE 38.97) underperforming the ‘LLM + 10 Topics’ oracle (IE 40.83) on Dragonball. The claim requires clarification: does ‘strongest non-LLM baseline’ refer to SAKI-RAG (IE 32.90) or a different aggregate? The specific baseline for the 8.24% figure must be explicitly named to avoid misrepresentation.

- **[science]** Section 3.3 states training uses ‘Qwen3-Embedding-4B’ and ‘Qwen3-32B’. These model versions (Qwen3) are not yet publicly released as of the current date (2024/2025 context). Unless these are internal unreleased models, the citation or model name is likely hallucinated or premature. If these refer to Qwen2.5 or similar, the text must be corrected to match available public artifacts to ensure reproducibility.
- **[writing]** Table 1 lists ‘LLM + 10 Topics’ as a baseline with IE 40.83 on Dragonball, yet the text describes MCompassRAG as ‘closely approaching’ this oracle. However, the table also lists ‘LLM’ (without topics) at 34.73. The distinction between the ‘LLM’ baseline and the ‘LLM + 10 Topics’ oracle is critical; the paper must clarify if the ‘LLM’ baseline includes topic metadata or if the ‘LLM + 10 Topics’ row represents a theoretical upper bound not achieved by any standard LLM baseline.

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_figure_critic.md

Figure 1

Figure 1 is a schematic overview of the pipeline, but it lacks a functional caption and contains a placeholder text error in the title area.

Figure 2

The figure provides a clear visual pipeline of the system architecture, but the caption is grammatically incomplete and fails to explicitly define the trainability icons (fire and snowflake) shown in the diagram.

Figure 3

The figure clearly presents the ablation study results with consistent formatting across panels, but the y-axis metric ‘IE’ is undefined and the x-axis label ‘Topics’ is slightly ambiguous regarding whether it represents a count or ID.

Figure 4

Figure 4 effectively illustrates the qualitative retrieval comparison between dense retrieval and MCompassRAG. The layout is clear, with distinct sections for retrieval candidates, dense retrieval scores, and the proposed method’s topic signals and MLP scores, all of which align perfectly with the detailed caption.

Figure 5

The figure effectively visualizes the separation of the gold chunk from distractors in the topic-enriched space, but the caption contains a significant truncation error and a typo.

Action items.

- **[writing]** Figure 1: The caption is explicitly marked as ‘(no caption)’, leaving the diagram’s components and flow unexplained.
- **[writing]** Figure 1: The diagram contains a placeholder text ‘Overview of .’ where the model name or system description should be.
- **[writing]** Figure 2: The caption contains multiple grammatical gaps where the model name ‘MCompassRAG’ is missing (e.g., ‘Overview of .’, ‘At inference time, selects...’). Additionally, the caption states that icons indicate trainability (fire for trained, snowflake for frozen), but the rendered figure lacks a legend or key to define these symbols.
- **[writing]** Figure 3: The x-axis label ‘Topics’ is ambiguous; the caption specifies ‘number of topics’, but the axis ticks (2, 4, ..., 20) are not explicitly labeled as counts, which could be confused with topic IDs.
- **[writing]** Figure 3: The y-axis label ‘IE’ is undefined in the figure and caption; the metric (e.g., Inverse Entropy, Information Efficiency) should be spelled out for clarity.
- **[writing]** Figure 5 caption: The sentence ends abruptly with ‘query–gold align’, indicating a truncation error.
- **[writing]** Figure 5 caption: ‘shareholde structure’ contains a typo (missing ‘r’).

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_jargon_police.md

The manuscript relies heavily on domain-specific jargon and undefined acronyms that hinder accessibility for non-specialist readers. The most critical issue is the introduction of “Information Efficiency (IE)” in the Abstract and Introduction without a definition. Readers cannot evaluate the claim of an “8.24% improvement” without knowing if this metric measures precision, recall,

token cost, or a composite score. This term must be explicitly defined in plain English upon first use.

Additionally, the acronym “RAG” (retrieval-augmented generation) appears in the Abstract and Introduction without expansion. Given the paper’s goal of broad applicability, standard acronyms should be spelled out at their first occurrence. The term “metadata bank” (Section 3.1) is used as a specific technical component but lacks a simple descriptive definition; it should be clarified as a “cached lookup table” or similar plain-language equivalent.

In Section 3.2, the phrase “student MLP” assumes the reader understands the “student-teacher” distillation paradigm and the specific architecture (Multi-Layer Perceptron). While common in ML, a brief parenthetical explanation (e.g., “a lightweight Multi-Layer Perceptron student model”) would improve clarity. Finally, the use of “orthogonal” in the Related Work section to describe the method’s relationship to query expansion is a mathematical metaphor that may be opaque to general readers; “complementary” or “independent” would be more accessible. These changes are necessary to ensure the paper’s contributions are understood by the intended broad audience.

Action items.

- **[writing]** The manuscript relies heavily on domain-specific jargon and undefined acronyms that hinder accessibility for non-specialist readers. The most critical issue is the introduction of “Information Efficiency (IE)” in the Abstract and Introduction without a definition. Readers cannot evaluate the claim of an “8.24% improvement” without knowing if this metric measures precision, recall, token cost, or a composite score. This term must be explicitly defined in plain English upon first use. Additionally

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_logical_consistency.md

The paper presents a coherent argument for using topic metadata to guide retrieval, but several logical gaps exist between the stated claims and the presented evidence.

First, the central claim of an “8.24% average improvement in Information Efficiency (IE)” over the “strongest non-LLM baseline” (Abstract, Introduction) requires precise definition. In Table 1, MCompassRAG (IE 38.97) is compared against SAKI-RAG (IE 32.90) and the “LLM + 10 Topics” oracle (IE 40.83). If the 8.24% figure is an average across six benchmarks, the specific baseline for each benchmark must be consistent. The text implies a comparison against non-LLM methods, yet the “LLM + 10 Topics” row is often the highest performer. The logic of the claim is ambiguous: is the improvement over the best *non-LLM* method, or is it a weighted average that includes the oracle? If the latter, the claim “over the strongest non-LLM baseline” is misleading if the oracle is significantly higher.

Second, the training mechanism described in Section 3.3 (“Training with LLM-Teacher

Distillation”) lacks a clear causal link. The authors state the LLM teacher labels relevance for *expanded* queries, while the student learns from *base* queries. The logical leap here is that the relevance signal derived from an expanded query (which contains more context) is directly transferable to a base query (which is shorter) without explicit alignment. The paper asserts this works via “distillation” but does not explain the mechanism by which the student learns to map base queries to the expanded query’s relevance distribution. Without this explanation, the claim that the student “avoids inference-time LLM calls” while maintaining high performance is not fully supported by the described training logic.

Third, the latency claim in the Conclusion (“5x lower latency than LLM-based baselines”) is not fully supported by the data in Table 2. The table compares MCompassRAG (174ms) against SAKI-RAG (925ms) and PageIndex (4408ms). While SAKI-RAG is a strong baseline, the “LLM” baseline mentioned in the text (likely referring to a full generative RAG pipeline) is not explicitly listed with a latency value in Table 2. If the “LLM” row in Table 1 corresponds to a different latency profile than SAKI-RAG, the 5x claim is an extrapolation not directly grounded in the provided table. The authors should explicitly list the latency of the “LLM” baseline to validate the 5x reduction claim.

Finally, the ablation study in Table 3 shows that removing the “Selection Policy” or “Abstraction” module lowers performance. However, the text in Section 5 states that “IE peaks at 12–15 topics,” while the main experiments use $K = 100$ topics (Section 4.1). This is a logical inconsistency: if performance peaks at 12–15 topics, why is the main configuration set to 100? The text in Section 5 mentions “more topics introduce noise,” which contradicts the choice of $K = 100$ unless the “12–15” refers to the *selected* topics (L or M) rather than the total topic count (K). The distinction between total topics (K) and selected topics (L, M) must be clarified to ensure the ablation results logically support the main configuration.

Action items.

- **[writing]** The paper presents a coherent argument for using topic metadata to guide retrieval, but several logical gaps exist between the stated claims and the presented evidence. First, the central claim of an “8.24% average improvement in Information Efficiency (IE)” over the “strongest non-LLM baseline” (Abstract, Introduction) requires precise definition. In Table 1, MCompassRAG (IE 38.97) is compared against SAKI-RAG (IE 32.90) and the “LLM + 10 Topics” oracle (IE 40.83). If the 8.24% figure is an ave

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_overreach.md

The paper makes several quantitative claims that slightly overreach the presented evidence, primarily regarding the magnitude of improvement over “strongest” baselines and the proximity to the oracle.

First, the abstract and introduction claim an average Information Efficiency (IE) improvement of **8.24%** over the “strongest efficient RAG baselines.” However, Table 1 (e000) and the detailed tables in the appendix (e001) reveal that the “LLM + 10 Topics” configuration consistently outperforms MCompassRAG across all datasets (e.g., IE 40.83 vs. 38.97 on Dragonball). The “LLM + 10 Topics” baseline is arguably the strongest efficient baseline presented, as it uses the same topic metadata but leverages an LLM for scoring. The 8.24% figure appears to be calculated against non-LLM baselines like SAKI-RAG or RAPTOR. The text must explicitly clarify that the improvement is relative to *non-LLM* baselines to avoid the implication that MCompassRAG outperforms the LLM-guided oracle.

Second, the claim of “**over 5x lower latency**” than the strongest baselines (Abstract, Section 1) is not uniformly supported. Table 2 (e000) shows MCompassRAG (174ms) is 5.3x faster than SAKI-RAG (925ms) but only 4.2x faster than REFRAG (720ms). While “over 5x” is true for one baseline, it is not a generalizable claim for the “strongest efficient baselines” as a group. The phrasing should be qualified (e.g., “up to 5x faster” or “5x faster than SAKI-RAG”) to maintain scientific precision.

Finally, the assertion in Section 4.2 that MCompassRAG “**closely approaches**” the LLM oracle is qualitatively supported by the small numerical gaps (e.g., 94.13 vs. 94.67 on SCI-DOCS), but the paper lacks a statistical analysis to confirm that this gap is negligible. Without a significance test or a discussion on the practical irrelevance of the remaining gap, the claim of “closely approaching” risks over-interpreting the data. The authors should either add a statistical validation or temper the language to “approaches within X%.”

These are minor over-claims that can be resolved by tightening the comparative language and clarifying the specific baselines used for the reported percentages.

Action items.

- **[writing]** The paper makes several quantitative claims that slightly overreach the presented evidence, primarily regarding the magnitude of improvement over “strongest” baselines and the proximity to the oracle. First, the abstract and introduction claim an average Information Efficiency (IE) improvement of 8.24% over the “strongest efficient RAG baselines.” However, Table 1 (e000) and the detailed tables in the appendix (e001) reveal that the “LLM + 10 Topics” configuration consistently outperforms MCompa

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_safety_ethics.md

The manuscript presents a novel retrieval framework, MCompassRAG, which utilizes topic metadata to guide retrieval without inference-time LLM calls. From a safety and ethics perspective, the paper is generally sound but lacks specific disclosures regarding data handling and potential bias.

Data Privacy and Third-Party API Usage: The methodology (Section 3.3, “Training with LLM-Teacher Distillation”) explicitly states the use of GPT-4o to synthesize training data (queries and relevance labels). While the paper notes the use of public benchmarks (e.g., HotpotQA, SQuAD), it does not address the data privacy implications of sending chunks from these corpora—some of which may contain sensitive or proprietary information (e.g., LegalBench-RAG, DRBench)—to an external API. The authors should clarify in the “Experimental Setup” or “Limitations” section that no Personally Identifiable Information (PII) or sensitive proprietary data was included in the prompts sent to OpenAI, or confirm that the data was anonymized prior to API usage.

Bias and Fairness: The “Limitations” section (Section 6) discusses technical constraints such as topic model quality and hyperparameter sensitivity but does not address algorithmic bias. Topic modeling and distillation processes can inadvertently amplify biases present in the source corpora (e.g., demographic biases in news data or legal biases in contract data). The authors should add a brief discussion on whether they evaluated the system for bias across different domains or user groups and how the topic-guided mechanism might impact fairness compared to standard dense retrieval.

Dual-Use and Misuse Potential: The system is designed to improve retrieval efficiency, which is generally positive. However, the ability to rapidly retrieve and synthesize information from large, potentially sensitive corpora (like legal or financial documents) could be misused for automated surveillance or generating misleading summaries if the retrieval is not carefully constrained. While the paper focuses on performance, a brief acknowledgment of the responsible deployment of such high-efficiency retrieval systems in sensitive domains would strengthen the ethical standing of the work.

Conclusion: The paper does not present immediate fatal safety risks, but the lack of transparency regarding data privacy in the distillation pipeline and the omission of bias considerations require minor revisions to meet standard ethical review criteria for AI research.

Action items.

- **[writing]** The training pipeline relies on GPT-4o (openai2024gpt4o) to generate synthetic queries and relevance labels (Section 3.3, Appendix A). The manuscript must explicitly state the data privacy policy regarding any proprietary or sensitive data potentially present in the target corpora (e.g., LegalBench-RAG, DRBench) when sent to the external LLM API, and confirm that no Personally Identifiable Information (PII) was included in the prompts.
- **[writing]** The ‘Limitations’ section (Section 6) acknowledges sensitivity to topic count and domain shifts but omits discussion of potential bias amplification. The authors should add a brief statement addressing whether the topic modeling or distillation process could inadvertently reinforce biases present in the training data (e.g., in LegalBench-RAG or HotpotQA) and how this was mitigated.
- **[writing]** The paper claims ‘over 5x lower latency’ than LLM-based baselines (Abstract, Section 1). While this is an efficiency claim, it has safety implications for deployment in high-stakes domains (e.g., legal/medical RAG) where speed might encourage over-reliance on potentially hallucinated outputs. The authors should briefly discuss the trade-off between

the efficiency gains and the risk of reduced interpretability compared to full LLM reasoning.

Scientific Evidence

Weighs the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_scientific_evidence.md

The paper presents a compelling architecture for topic-guided retrieval, but the scientific evidence supporting the central quantitative claims requires clarification to ensure robustness.

First, the headline claim of an “8.24% average IE improvement” (Abstract, Introduction) is ambiguous regarding the baseline. Table 1 (e000) demonstrates that MCompassRAG consistently underperforms the “LLM + 10 Topics” oracle across all datasets (e.g., 38.97 vs. 40.83 IE on Dragonball). The 8.24% figure likely represents the gain over the strongest *non-LLM* baseline (SAKI-RAG), but the text does not explicitly define this comparison group. Without this explicit definition, the claim risks misinterpretation as a gain over the best available method. The authors must specify the exact baseline used for this aggregate statistic.

Second, the latency claim of “over 5x lower latency” (Abstract) appears to rely on a specific comparison to SAKI-RAG (174ms vs. 925ms in Table 2, e000). However, the paper also compares against PageIndex (4408ms), where the speedup is ~25x. The phrasing “strongest efficient RAG baselines” is vague; if the claim is intended to highlight efficiency against the *most* efficient baseline, the 5x figure is misleadingly low. If it is against the *strongest* baseline (which might be PageIndex in terms of performance), the comparison is inconsistent. The authors should clarify which baseline justifies the “5x” figure or report the range of speedups observed.

Finally, the training methodology relies heavily on LLM-teacher distillation using GPT-4o (Section 3.3). The paper does not report any metrics regarding the quality, consistency, or potential bias of these synthetic labels. In distillation pipelines, teacher errors can propagate and limit the student’s ceiling. A brief discussion on the reliability of the teacher labels or an analysis of the student-teacher gap (beyond the IE plots) would strengthen the validity of the training evidence.

While the ablation studies (Section 5) effectively demonstrate the contribution of individual components (selection, abstraction), the primary performance claims need tighter grounding in the reported tables to avoid overstatement.

Action items.

- **[science]** The claim of ‘8.24% average IE improvement’ (Abstract, Intro) lacks a defined baseline. Table 1 shows MCompassRAG trailing the ‘LLM + 10 Topics’ oracle on all metrics. Clarify if the 8.24% is against the strongest non-LLM baseline (SAKI-RAG) or a different aggregate, and report the specific baseline used for this calculation to avoid ambiguity.
- **[science]** The latency claim of ‘5x lower latency’ (Abstract) is not supported by the provided data. Table 2 shows MCompassRAG at 174ms vs. SAKI-RAG at 925ms (~5.3x), but PageIndex

is 4408ms (~25x). The text implies a comparison to ‘strongest efficient RAG baselines’ but does not explicitly state which baseline yields the 5x figure, risking cherry-picking.

- **[science]** The training data synthesis relies on GPT-4o (Section 3.3) without reporting the inter-annotator agreement or consistency of the teacher labels. Given the reliance on distillation, the potential for teacher bias or hallucination in the relevance labels is a significant threat to validity that requires a brief discussion or error analysis.

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_statistical_analysis.md

The manuscript presents a novel retrieval framework, MCompassRAG, but the statistical rigor supporting its central claims is currently insufficient. While the authors report substantial improvements in Information Efficiency (IE) and latency, the analysis relies almost exclusively on point estimates without measures of variance or statistical significance testing.

Specifically, the claim that the method “consistently outperforms all baselines” (Section 4.2) is not statistically substantiated. For instance, Table 1 reports an average IE improvement of 8.24% over the strongest non-LLM baseline. However, there is no indication of whether this difference is statistically significant across the six benchmarks. Given the variability inherent in retrieval tasks and the specific dataset splits used, a paired t-test or bootstrap confidence intervals (e.g., 95% CI) are required to validate that these gains are not due to random chance.

Furthermore, the comparison against the “LLM + 10 Topics” oracle (Table 1 and Table 2) lacks context regarding variance. If the oracle results are single-point measurements (common in LLM-based baselines due to cost), comparing them against a student model without error bars creates an unfair or misleading statistical comparison. The ablation study in Table 3 (Section 5) also presents marginal gains (e.g., IE dropping from 38.97 to 38.03 when removing abstraction) without reporting standard deviations. It is impossible to determine if these component contributions are statistically distinguishable from zero.

Finally, the hyperparameter sensitivity analysis (Section 5, Figure 4) notes that performance “peaks at 12–15 topics” (text) but Table 4 shows $K = 100$ as the optimal setting. This discrepancy in the reported optimal topic count (K) needs clarification, and the performance drop at higher K values should be accompanied by significance tests to confirm the degradation is real and not noise. The authors must provide statistical validation (p-values, CIs, or standard deviations) for all comparative claims to support the paper’s conclusions.

Action items.

- **[science]** Report statistical significance (e.g., paired t-tests or bootstrap CIs) for the claimed 8.24% average IE improvement and latency reductions across the six benchmarks. Point estimates alone are insufficient to support the claim of consistent outperformance.

- **[science]** Clarify the experimental design for the ‘LLM + 10 Topics’ oracle baseline. Is this a single run or an average? If single, the comparison against the student model (which likely has variance) is statistically invalid without error bars or significance testing.
- **[science]** The ablation study (Table 3) shows small absolute differences (e.g., IE 38.97 vs 38.03). Without reported standard deviations or p-values, it is unclear if the ‘Abstraction’ and ‘Selection’ components provide statistically significant gains over the noise floor.

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_writing_quality.md

The manuscript presents a clear and generally well-structured argument for the proposed MCompassRAG framework. The flow from the problem statement (granularity trade-off) to the solution (topic metadata as a compass) is logical and easy to follow. The abstract effectively summarizes the core contribution and results.

However, there are specific writing issues that detract from the professional polish of the paper. Most notably, the title and abstract rely on the LaTeX placeholder `\name{}` rather than explicitly stating “MCompassRAG”. In a final submission, the title must be self-explanatory, and the abstract should not contain undefined macros. This is a critical readability issue for any reader not compiling the LaTeX source.

Additionally, there are minor typographical errors, such as the trailing space in the section heading “Semantic Guidance and Efficient Retrieval. “ in Section 2. While the prose is generally concise, there are occasional instances where the distinction between “topic-level signals” and “topic metadata” could be tightened to ensure terminological consistency. The qualitative analysis section is well-written but relies heavily on figure references; ensuring the text stands alone as much as possible would improve accessibility.

Overall, the writing quality is high, but the presence of the placeholder macro and minor typos necessitates a minor revision to ensure the paper is ready for publication.

Action items.

- **[writing]** The title and abstract use the placeholder ‘`\name{}`’ instead of the actual model name ‘MCompassRAG’. This must be replaced with the full name or a defined macro before submission to ensure the paper is self-contained and readable.
- **[writing]** In Section 2 (Related Work), the second paragraph heading ‘Semantic Guidance and Efficient Retrieval. ‘ contains a trailing space before the closing brace. While minor, this is a typographical error that should be cleaned up for professional presentation.
- **[writing]** The phrase ‘Topic Metadata as a Semantic Compass’ is used as a metaphor, but the text occasionally shifts between ‘topic-level signals’ and ‘topic metadata’ without clear distinction. Ensure consistent terminology throughout the introduction and method sections.

to avoid reader confusion.